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Genomics Bioinformatics Software Engineer

UK Health Security Agency

Apply before 11:55 pm on Wednesday 11th November 2026

UK Health Security Agency

Reference number

486947

Salary

£56,185 - £70,566

per annum, pro rata

A Civil Service Pension with an employer contribution of 28.97%

Job grade

Grade 7

Contract type

Fixed term
Loan
Secondment

Length of employment

24 months from start date

Type of role

Information Technology
Science
Other

Working pattern

Flexible working, Full-time, Job share, Part-time

Number of jobs available

1

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Location

Birmingham, Leeds, Liverpool, London Canary Wharf (Core HQs)

About the job

Job summary

The genomics team support the delivery of the UKHSA Pathogen Genomics service which comprise of a high throughput, distributed sequencing capability. The sequencing capability was developed to support the COVID pandemic response and delivery of the pathogen genomics programme.

The work of the Pathogen Genomics service is to support UKHSA s genomics surveillance programmes, to aid detection of outbreaks and transmission of infections, to characterise and correlate genomics to virulence and antimicrobial resistance mechanisms, and generate evidence to support effective interventions, all within the wider UKHSA objective, to significantly reduce their overall public health burden.

Job description

Contribute to the development and implementation of a strategy, jointly with partners, for creating software processes for WGS processing, analysis, reporting and storage within UKHSA.

To design and deploy processes to collect and report business intelligence information in real-time using SQL databases and dashboarding tools.

To support delivery of routine WGS-based services including managing issues and tickets raised by customers.

To coordinate the UKHSA contribution to development of new software and design with appropriate architecture for more efficient and scalable ways of automated processing, analysis, interpretation and visualisation of sequence and epidemiological data.

To work with bioinformatics colleagues to migrate, as appropriate, current analytical pipelines to the new architecture for software solutions.

To ensure best practice for software development, testing and implementation is followed in the WGS Development and Implementation Service and across UKHSA.

To ensure compliance with quality procedures and achieve appropriately recognised software accreditation

To support software improvements required to enhance current UKHSA service delivery.

Detailed job description and main responsibilities

The role is National and may involve travel to UKHSA locations in England for some meetings and site visits.

The above is only an outline of the tasks, responsibilities and outcomes required of the role. You will carry out any other duties as may reasonably be required by your line manager.

The job description and person specification may be reviewed on an ongoing basis in accordance with the changing needs of the division and UKHSA.

It should be noted that the work of the division is of a confidential nature and must not be communicated to other persons except where required for authorised purposes.

PROFESSIONAL DEVELOPMENT

Identify, discuss and action own professional performance and training / development needs with your line manager through appraisal / individual development plan. Attending internal / external training events

To participate in all mandatory training as required, i.e., fire safety, information governance and all other mandatory training.

KEY WORKING RELATIONSHIPS

The post holder will develop working relationships and communicate regularly with a wide range of individuals, clinical and non-clinical, internal and external to UKHSA. This will include.

Internal:

Members of Pathogen Genomics Teams

Laboratory staff across the UKHSA

ICT & Technology teams

Members of Laboratory Quality Manager Teams

Staff within Laboratory Informatics Team

External:

LIMS vendors

NHS Trust ICT & Technology teams

External stakeholders including customers and diverse external institutions

National and international regulatory authorities

We are looking for a capable, highly motivated software developer with experience in designing / interacting with genomic software solutions (bioinformatics pipelines).

The priority is to deliver on the strategic aim of developing unified informatics solutions for assembling, analysing and reporting WGS results. This needs to be fit-for-purpose for a wide range of bacterial, viral and fungal pathogens and builds on extensive work establishing bespoke and successful routine WGS-based services for several key infectious disease agents. Solutions will need to be scalable and cope with high-throughput sequence processing yielding actionable information for public health and patient centred management. Solutions should also collect and report business intelligence information in real-time using SQL databases and dashboarding tools.

We require interfaces using modern web technologies & frameworks, so we have a broad range of software development challenges for the right person. A key activity is the establishment and deployment of robust and often novel analytical pipelines developed to a standard, which will meet necessary ISO standards and achieve UKAS accreditation.

The successful candidate will have a wide range of skills and an interest in developing applications for processing genomic data. Most of the systems to date are written in Python, however, we recognize potential advantages in migrating to C#, C++, Java and F# and require skills in these areas.

The software needs to be fully documented, modular, version-controlled and incorporated into a workflow with containerization. The software will need to be flexibly developed so it can be variably hosted on a range of computational environments including UKHSA s high performance computer cluster, EBI and commercial cloud providers such as Amazon or Azure.

The software will need to be developed in line with the needs of users and have applications to not only support UKHSA, but partner users such as the Devolved Administrations, Animal and Plant Agency and international users.

Implement software solutions for visualization of genomic analyses linked to epidemiological and geospatial data for the better management of outbreaks.

Dissemination of software code will be through UKHSA GitLab and GitHub repositories Provision of software engineering support and training across the Agency and beyond. The division is committed to working flexibly and collaboratively as part of a close team and will ensure the successful individual is supported to deliver challenging tasks that will enable them to progress within their career.

This list is not exhaustive.

Person specification

Essential criteria:

- Higher degree (i.e. MSc or above or equivalent qualification) in Computer Science or related discipline
- Extensive experience of designing, developing and delivering large and complex software solutions
- Experience of distributed computer resources e.g., Docker, Singularity
- Demonstrable skills in developing user interfaces for complex software systems
- Proven ability to write high quality software documentation and implementation of quality standards for coding among a team.
- Expert knowledge/experience of Unix, PostgreSQL or similar RDBMS. Knowledge of Cassandra or similar NoSQL Database architectures and

Proven expert skills in at least one object-oriented programming language, preferably Java or C++

Proven skills in developing and modifying code used for data analysis

Benefits

Alongside your salary of £56,185, UK Health Security Agency contributes £16,276 towards you being a member of the Civil Service Defined Benefit Pension scheme. Find out what benefits a Civil Service Pension provides (opens in a new window).

Learning and development tailored to your role

An environment with flexible working options

A culture encouraging inclusion and diversity

A Civil Service pension with an employer contribution of 28.97%

We pride ourselves as being an employer of choice, where Everyone Matters promoting equality of opportunity to actively encourage applications from everyone, including groups currently underrepresented in our workforce.

UKHSA ethos is to be an inclusive organisation for all our staff and stakeholders. To create, nurture and sustain an inclusive culture, where differences drive innovative solutions to meet the needs of our workforce and wider communities. We do this through celebrating and protecting differences by removing barriers and promoting equity and equality of opportunity for all.

Things you need to know

Artificial intelligence

Artificial intelligence can be a useful tool to support your application, however, all examples and statements provided must be truthful, factually accurate and taken directly from your own experience. Where plagiarism has been identified (presenting the ideas and experiences of others, or generated by artificial intelligence, as your own) applications may be withdrawn and internal candidates may be subject to disciplinary action. Please see our

candidate guidance (opens in a new window) for more information on appropriate and inappropriate use.

Selection process details

This vacancy will be assessed using a competency-based framework which will assess your qualifications, knowledge and experience and / or skills and abilities outlined in the essential criteria.

Stage 1: Application & Sift
Competency Based

You will be required to complete an application form. You will be assessed on the listed 7 essential criteria, and this will be in the form of a:

- Application form (Employer/ Activity history section on the application)
- 1000 word supporting statement.

This should outline how you consider your skills, experience and knowledge provide evidence of your suitability for the role, with reference to the essential criteria.

You will receive a joint score for your application form and statement. (The application form is the kind of information you would put into your C.V please be advised you will not be able to upload your CV. Please complete the application form in as much detail as possible). **Please do not email us your CV.**

Healthjobs UK has a word limit of 1500, but your supporting statement must be no more than 1000 words. We will not consider any words over 1000 words.

Longlisting:

In the event of a large number of applications, we will longlist into 3 piles of:

- Meets all essential criteria
- Meets some essential criteria
- Meets no essential criteria

If used, the pile(s) Meets all essential criteria will proceed to shortlisting.

Shortlisting:

In the event of a large number of applications we will conduct an initial sift, on the lead criteria of:

- Extensive experience of designing, developing and delivering large and complex software solutions
- Demonstrable skills in developing user interfaces for complex software systems
- Expert knowledge/experience of Unix, PostgreSQL or similar RDBMS. Knowledge of Cassandra or similar NoSQL Database architectures and Proven expert skills in at least one object-oriented programming language, preferably Java or C++

If you are successful at this stage, you will progress to interview & assessment.

Feedback will not be provided at this stage.

Stage 2: Interview
Competency Based

You will be invited to a single remote interview.

Knowledge, experience, skills & abilities will be tested at interview.

UKHSA are working to develop modern enterprise solutions for storing Sanger and NGS sequence information in ways that enable laboratory staff to view, query and analyse the data.

You will be required to prepare a 10 minute presentation highlighting the user and technical factors that should be considered when developing such a solution.

Include/describe how your skills and experience would support the successful design, development and implementation of such a solution.

Interview dates are yet to be confirmed.

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Location

This role is being offered as hybrid working based at any of our Core HQs. We offer great flexible working opportunities at UKHSA and operate using a hybrid working model where business needs allow. This provides us with greater flexibility about how and where we work, to get the best from our workforce. As a hybrid worker, you will be expected to spend a minimum of 60% of your contractual working hours (approximately 3 days a week pro rata, averaged over a month) working at one of UKHSA's core HQs (Birmingham, Leeds, Liverpool, and London). Our core HQ offices are modern and newly refurbished with excellent city centre transport links and benefit from co-location with other government departments such as the Department for Health and Social Care (DHSC).

Security Clearance Level Requirement

Successful candidates must pass a basic disclosure and barring security check before they can be appointed.

Successful candidates must meet the security requirements before they can be appointed. The level of security needed is Basic Personnel Security Standard

Eligibility Criteria

External: This vacancy is open to all external applicants (anyone) from outside the Civil Service as well as internal applicants.

If this role is fixed term and you are not currently a Civil Servant, you will take up the post on a Fixed Term appointment; you may be able to take this role up as a Secondment. If you are an existing Civil Servant, based outside of UKHSA, you will take up the post as a loan which you will need your department to agree. Loans are normally agreed for a period of two years. However, this may be extended where required, subject to agreement between UKHSA and your home department. You cannot take the post up as

a fixed term. If you are an existing UKHSA member of staff, you will take up the post as either a level transfer or a temporary promotion as per the UKHSA s Pay policy.

If this role is permanent, it may also be available as a loan (for existing Civil Servants only) or secondment arrangement, subject to business need and agreement from the hiring manager and, where applicable, the employee's home organisation.

Reserve List

If more than the required number of suitable candidates pass the interview criteria, you may be kept on a reserve list for 12 months subject to your agreement. You may be contacted, in merit-order, if similar roles with closely matching essential criteria become available and the department choose to appoint from a reserve list.The panel will assess if candidates meet the requirement of the role first, using a specific benchmark system. If you are interviewed for the post and do not meet the required threshold for the specified grade, your application may be assessed against a similar, lower grade role and you may be offered the post should one be available.

Incident Response Activity

From time to time, the role holder may be required to support UKHSA s incident response activity by contributing their appropriate expertise within an Incident Cell; during such periods, supported by the line manager, this work will take priority over normal day-to-day responsibilities, commensurate with their grade, skills and experience.

Future location

UKHSA is investing in a new state-of-the-art National Biosecurity Centre in Harlow, Essex, which will eventually bring together teams currently based at Canary Wharf, Colindale and Porton Down. For more details, please see: Huge biosecurity centre investment to boost pandemic protection - GOV.UK.

The new facilities will start becoming operational in the mid-2030s, with full completion by 2038. Staff will move in phases as facilities become available. If you're appointed to a role currently based at Canary Wharf, Colindale or Porton Down, please note that we'll continue investing in these sites for the

next decade. As we get closer to the transition, we'll provide full information about relocation support available to staff.

Reasonable Adjustments

The Civil Service is committed to making sure that our selection methods are fair to everyone. To help you during the recruitment process, we will consider any reasonable adjustments that could help you. An adjustment is a change to the recruitment process or an adjustment at work. This is separate to the Disability Confident Scheme. If you need an adjustment to be made at any point during the recruitment process you should contact the recruitment team in confidence as soon as possible to discuss your needs.

You can find out more information about reasonable adjustments across the Civil Service here: https://www.civil-service-careers.gov.uk/reasonable-adjustments/

International Police check

If you have spent more than 6 months abroad over the last 3 years you may need an International Police Check. This would not necessarily have to be in a single block, and it could be time accrued over that period.

Internal Fraud check

If successful for this role as one aspect of pre-employment screening, applicant s personal details name, national insurance number and date of birth - will be checked against the Cabinet Office Internal Fraud Hub and anyone included on the database will be refused employment unless they can show exceptional circumstances. Currently this is only for External candidates to the Civil Service.

Careers website

Please visit our careers site for more information https://gov.uk/ukhsa/careers

Please note: The Digital and Data Directorate, where this role sits, is reviewing its Target Operating Model. While the process is still in the planning stage, some roles may be affected by future changes. We are committed to

keeping candidates informed and will share updates as they become available.

Salary Information

If you are successful at interview, and are moving from another government department, NHS, or Local Authority, the relevant starting salary principles for level transfers or promotions will apply. Otherwise, roles are offered at the pay scale minimum for the grade, but in exceptional circumstances there may be flexibility if you are able to demonstrate you are already in receipt of an existing, higher salary. Pay increases are through the relevant annual pay award for the role and terms.

Please be aware that the salary is based on the office location.

Grade 7

£56,185- £66,581 (National)

£58,340- £68,574 (Outer London)

£60,494- £70,566 (Inner London)

Feedback will only be provided if you attend an interview or assessment.

Security

Successful candidates must undergo a basic (or equivalent) criminal record check.

People working with government assets must complete baseline personnel security standard (opens in new window) checks.

Nationality requirements

This job is broadly open to the following groups:

UK nationals

nationals of the Republic of Ireland

nationals of Commonwealth countries who have the right to work in the UK

- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities with settled or pre-settled status under the European Union Settlement Scheme (EUSS) (opens in a new window)
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities who have made a valid application for settled or pre-settled status under the European Union Settlement Scheme (EUSS)
- individuals with limited leave to remain or indefinite leave to remain who were eligible to apply for EUSS on or before 31 December 2020
- Turkish nationals, and certain family members of Turkish nationals, who have accrued the right to work in the Civil Service

Further information on nationality requirements (opens in a new window)

Working for the Civil Service

The Civil Service Code (opens in a new window) sets out the standards of behaviour expected of civil servants.

We recruit by merit on the basis of fair and open competition, as outlined in the Civil Service Commission's recruitment principles (opens in a new window).

The Civil Service embraces diversity and promotes equal opportunities. As such, we run a Disability Confident Scheme (DCS) for candidates with disabilities who meet the minimum selection criteria.

Diversity and Inclusion

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan (opens in a new window) and the Civil Service Diversity and Inclusion Strategy (opens in a new window).

Apply and further information

This vacancy is part of the Great Place to Work for Veterans (opens in a new window) initiative.

The Civil Service welcomes applications from people who have recently left prison or have an unspent conviction. Read more about prison leaver recruitment (opens in new window).

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Contact point for applicants

Job contact :

Name : Nigel Poole

Email : Nigel.Poole@ukhsa.gov.uk

Recruitment team

Email : recruitment@ukhsa.gov.uk

Further information

The law requires that selection for appointment to the Civil Service is on merit on the basis of fair and open competition as outlined in the Civil Service Commission's Recruitment Principles. If you feel your application has not been treated in accordance with the Recruitment Principles, and you wish to make a complaint, in the first instance, you should contact UKHSA Public Accountability Unit via email: Complaints@ukhsa.gov.uk If you are not satisfied with the response you receive from the Department, you can contact the Civil Service Commission: Visit the Civil Service Commission website here.

Civil Service Commission

disability confident LEADER